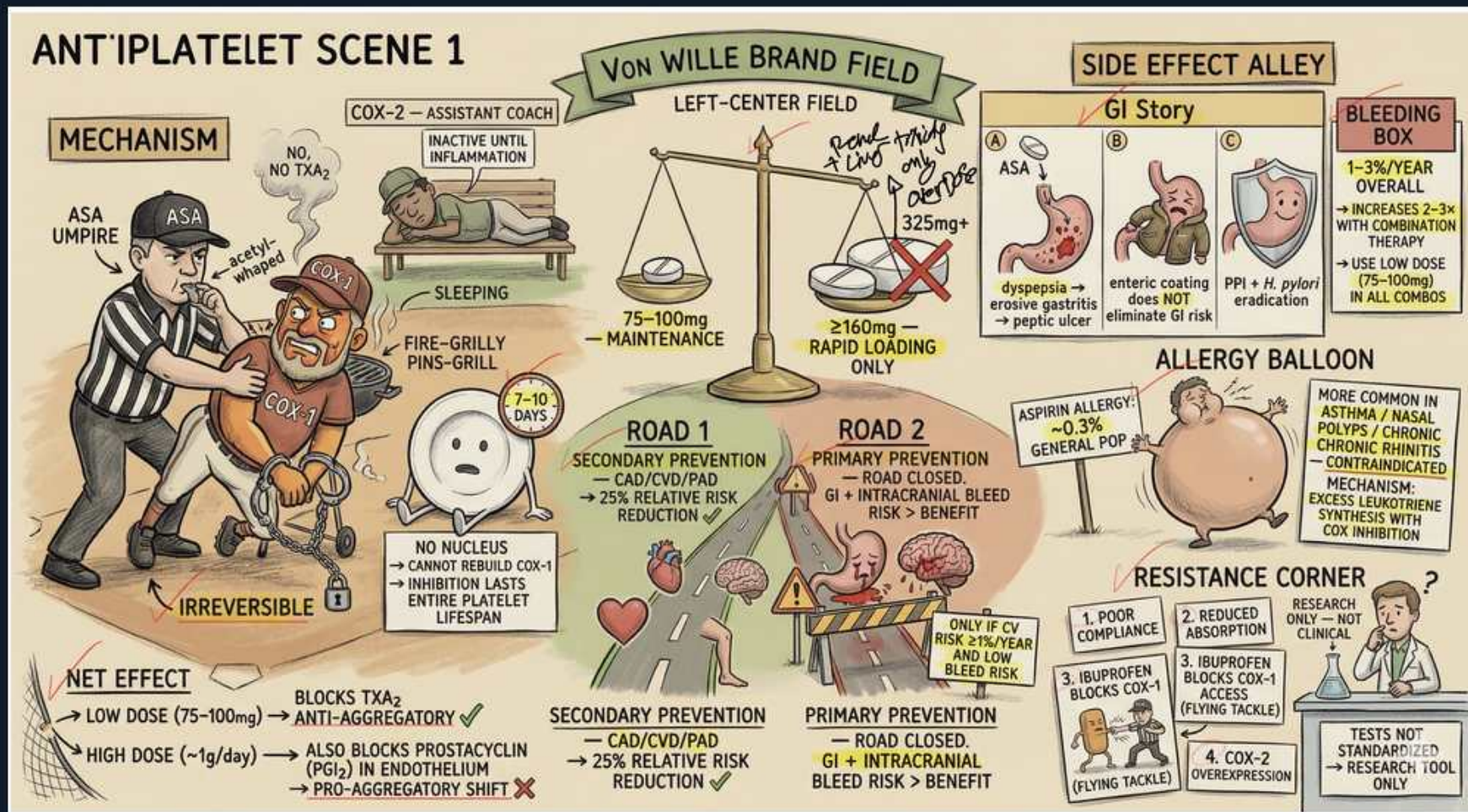


Antiplatelets — Aspirin (COX-1 Umpire)

ANTIPLATELET SCENE 1



1. ASA umpire acetylating COX-1

WHAT YOU SEE

An umpire labeled ASA pressing on a grill-master labeled COX-1, with the note 'acetyl-whaped'.

WHAT IT ENCODES

Aspirin (ASA) irreversibly acetylates serine 530 of COX-1, the enzyme that makes thromboxane A2 (TXA2). This blocks TXA2 synthesis and the platelet's amplification of aggregation. ASA is the prototype antiplatelet.

BOARD TRAP

Aspirin acetylation is IRREVERSIBLE and covalent — not competitive. Ibuprofen is reversible and competes for the same channel, so taking ibuprofen before aspirin can block aspirin's antiplatelet effect.

2. COX-1 grilling TXA2

WHAT YOU SEE

COX-1 character at a fire-grill ('fire-grilly pins-grill') making thromboxane smoke.

WHAT IT ENCODES

COX-1 converts arachidonic acid to TXA2, a potent pro-aggregatory, vasoconstricting agonist. Aspirin's antiplatelet benefit comes from shutting down platelet COX-1 -> TXA2.

BOARD TRAP

TXA2 (platelet, pro-thrombotic) is balanced against endothelial PGI2 (anti-thrombotic). Low-dose aspirin selectively suppresses platelet TXA2 while sparing endothelial PGI2.

3. COX-2 assistant coach (sleeping)

WHAT YOU SEE

A second coach labeled COX-2 asleep on a bench, 'inactive until inflammation'.

WHAT IT ENCODES

COX-2 is largely inducible by inflammation and is not the main platelet enzyme; platelets rely on COX-1. This is why low-dose aspirin's antiplatelet effect is COX-1 driven.

BOARD TRAP

Selective COX-2 inhibitors (coxibs) do NOT give antiplatelet protection and actually raise cardiovascular thrombotic risk by suppressing endothelial PGI2 without blocking platelet TXA2 — a board favorite.

4. No nucleus -> 7-10 day inhibition

WHAT YOU SEE

A sad platelet with a padlock and box: 'no nucleus -> cannot rebuild COX-1 -> inhibition lasts entire platelet lifespan', clock '7-10 days'.

WHAT IT ENCODES

Platelets are anucleate and cannot resynthesize COX-1, so a single aspirin dose inhibits that platelet for its ~7-10 day lifespan. Functional recovery needs new platelets (~10-12%/day turnover).

BOARD TRAP

This is why aspirin is held ~5-7 days before surgery and why the antiplatelet effect persists long after the drug is cleared. Don't tie aspirin duration to its short plasma half-life.

5. Net effect: low vs high dose

WHAT YOU SEE

Net-effect box: low dose 75-100mg blocks TXA2 (anti-aggregatory, check) vs high dose ~1g/day also blocks PGI2 -> pro-aggregatory shift (X).

WHAT IT ENCODES

Low-dose ASA (75-100 mg) selectively inhibits platelet TXA2 = antiaggregatory. High dose (~1 g/day) also blocks endothelial prostacyclin (PGI2), shifting toward pro-aggregation and losing benefit.

BOARD TRAP

'More aspirin = more antiplatelet protection' is wrong. High dose blunts endothelial PGI2 (the aspirin/prostacyclin dilemma); the cardioprotective dose is LOW dose.

6. 75-100 mg maintenance dose

WHAT YOU SEE

A scale balancing a 75-100 mg tablet labeled MAINTENANCE.

WHAT IT ENCODES

75-100 mg daily is the chronic maintenance antiplatelet dose for secondary prevention. It maximizes TXA2 blockade while sparing PGI2 and minimizing GI/bleed risk.

BOARD TRAP

Maintenance (75-100 mg daily) is distinct from the loading dose. Confusing maintenance with the 162-325 mg loading dose for ACS is a dosing trap.

7. ≥160-325 mg rapid loading only

WHAT YOU SEE

A 325mg+ tablet with red X and label '≥160mg — RAPID LOADING ONLY', annotation 'renal+liver only / overdose'.

WHAT IT ENCODES

A loading dose of ~162-325 mg (chewed, non-enteric) is given once at ACS onset for rapid platelet inhibition, then dropped to low-dose maintenance. Chronic high dose is not beneficial.

BOARD TRAP

The high dose is for ACUTE LOADING only, then de-escalate. Continuing 325 mg chronically adds GI/bleeding risk without added benefit and risks the pro-aggregatory PGI2 shift.

8. Road 1 — secondary prevention benefit

WHAT YOU SEE

A green open road 'ROAD 1: SECONDARY PREVENTION — CAD/CVD/PAD -> 25% relative risk reduction'.

WHAT IT ENCODES

In established disease (prior MI, stroke, CAD, PAD), aspirin gives ~25% relative risk reduction in vascular events. Secondary prevention is the clear-benefit indication.

BOARD TRAP

Aspirin's strong evidence is for SECONDARY prevention. Don't generalize the 25% RRR to healthy primary-prevention patients, where benefit is marginal.

9. Road 2 — primary prevention closed

WHAT YOU SEE

A barricaded road 'ROAD 2: PRIMARY PREVENTION — ROAD CLOSED. GI + intracranial bleed risk > benefit', with 'only if CV risk $\geq 1\%$ /year and low bleed risk'.

WHAT IT ENCODES

For PRIMARY prevention, routine aspirin is no longer recommended because GI and intracranial bleeding risk outweighs benefit; consider only in select high-CV-risk, low-bleed patients.

BOARD TRAP

Modern guidelines REMOVED routine aspirin for primary prevention (esp. elderly). The trap is starting aspirin in a healthy patient 'for prevention' — bleeding risk wins.

10. GI Story — dyspepsia to PUD

WHAT YOU SEE

Side Effect Alley GI panel: A) dyspepsia/erosive gastritis -> peptic ulcer, B) enteric coating does NOT eliminate GI risk, C) PPI + H. pylori eradication.

WHAT IT ENCODES

Aspirin causes dose-dependent GI injury (dyspepsia, erosive gastritis, peptic ulcer) by blocking protective gastric COX-1 prostaglandins — a systemic, not just topical, effect. Manage with PPI and H. pylori eradication.

BOARD TRAP

Enteric coating does NOT eliminate GI/ulcer risk because injury is systemic (COX-1 mediated), not just local contact. Switching to enteric-coated to 'prevent ulcers' is the trap.

11. Bleeding Box — 1-3%/year

WHAT YOU SEE

Red BLEEDING BOX: 1-3%/year overall, increases 2-3x with combination therapy, use low dose in all combos.

WHAT IT ENCODES

Major bleeding risk on aspirin is roughly 1-3%/year, rising 2-3x with dual/combination antithrombotic therapy. Low dose is used in all combinations to limit bleeding.

BOARD TRAP

Adding a second antiplatelet or an anticoagulant multiplies bleeding risk — keep aspirin at LOW dose in any combo (DAPT or triple therapy).

12. Allergy Balloon — aspirin-exacerbated respiratory disease

WHAT YOU SEE

A swollen balloon person: aspirin allergy ~0.3% general pop; more common in asthma/nasal polyps/chronic rhinitis; mechanism = excess leukotriene synthesis with COX inhibition.

WHAT IT ENCODES

Aspirin hypersensitivity (~0.3% general; far higher in Samter's triad — asthma, nasal polyps, chronic rhinosinusitis) arises from COX-1 blockade shunting arachidonic acid to excess leukotrienes. Contraindicated in this triad.

BOARD TRAP

The reaction is pseudo-allergic (leukotriene shunting), not true IgE allergy, and it cross-reacts with ALL NSAIDs. Giving another NSAID to an aspirin-allergic asthmatic is the trap.

13. Resistance Corner

WHAT YOU SEE

Resistance Corner panel: 1) poor compliance, 2) reduced absorption, 3) ibuprofen blocks COX-1 access (flying tackle), 4) COX-2 overexpression; 'tests not standardized -> research tool only'.

WHAT IT ENCODES

'Aspirin resistance' (poor compliance, malabsorption, ibuprofen competition at COX-1, COX-2 overexpression) is a research concept; assays are not standardized and not used clinically.

BOARD TRAP

Don't order an 'aspirin resistance' test in practice — it is non-standardized/research only. Also: dose ibuprofen ≥ 2 h AFTER aspirin to avoid the COX-1 access blockade ('flying tackle').